Student Performance Analysis — Methodology and Findings

Personal Data Analytics Project | Microsoft Excel

1. Analytical Questions

I wanted to understand which factors were associated with students' final examination performance. I focused on five areas and used the following questions to guide my analysis:

- **Study Time: Do students who spend more hours studying tend to achieve higher final examination scores?**
- **Internet Access: Do students with internet access have different average examination scores from those without access?**
- **Part-Time Employment: Do students with part-time jobs have different average scores from students without jobs?**
- **Extracurricular Activities: Do students who participate in extracurricular activities perform differently from those who do not?**
- **Parental Education: Do average examination scores vary across different parental education levels?**

These questions helped me examine each factor separately and compare the results to understand which patterns were more noticeable in the dataset.

2. Data Understanding and Validation

I began by inspecting the dataset to understand its structure, the variables available, and their relevance to my analysis. The dataset contained 1,000 student records and 12 columns, including study time, attendance, sleep hours, internet access, parental education, part-time employment, previous grades, and final examination scores.

I checked for missing values, duplicate records, inconsistent values, inappropriate data types, invalid values, and potential outliers. My checks did not identify any of these issues in the dataset I analysed, so I did not identify a need for data cleaning based on those checks.

This initial review helped me establish that the dataset was suitable for proceeding with the planned analysis, while recognizing that these checks could not rule out every possible data-quality issue.

3. Analytical Approach

I used Microsoft Excel to organize, examine, and analyse the data. My work involved:

- **Descriptive statistics: I calculated summary measures to understand examination score distributions and overall performance.**
- **PivotTables: I summarized and compared examination scores across categories such as internet access, part-time employment, extracurricular activities, and parental education.**
- **Visualizations: I created histogram, scatter plot, and column charts to explore score distributions, relationships, and differences between groups.**
- **Interactive dashboard: I brought key metrics and visualizations together in an interactive dashboard to make the results easier to explore and understand.**

I then reviewed the results across the five focus areas rather than relying on one comparison to explain student performance.

4. Findings and Interpretation

My analysis produced different results across the factors I investigated:

- **Study Time: I observed a positive relationship between study time and final examination scores. However, students with similar study times did not always achieve the same scores, suggesting that study time alone did not explain all the variation in performance.**
- **Internet Access: Students with internet access had an average score of 84.32, compared with 79.29 for students without access. This was a difference of 5.03 points in the dataset.**
- **Part-Time Employment: Students without part-time jobs averaged 84.62, compared with 81.35 for students with jobs, a difference of 3.27 points.**
- **Extracurricular Activities: Students who participated in extracurricular activities averaged 83.78, compared with 83.32 for those who did not. The difference was relatively small at 0.46 points.**
- **Parental Education: Average examination scores varied only slightly across parental education levels, ranging from approximately 83.29 to 85.08.**

These results showed me that the size of the observed differences varied across the factors. Internet access and part-time employment showed more noticeable differences in average scores than extracurricular activities and parental education in this dataset. However, these comparisons do not establish that the factors caused the differences observed.

5. Recommendations

Based on my findings, I developed recommendations and recorded them in the Analysis & Recommendations worksheet in my Excel workbook.

The recommendations include encouraging effective study habits, considering access to reliable internet and digital learning resources, and exploring flexible academic support for students balancing part-time employment with their studies. I also recommended maintaining support for extracurricular participation and avoiding the use of parental education alone to determine a student's support needs.

These are potential areas for further investigation and support. Additional evidence would be needed to determine whether specific interventions would improve examination performance.

6. Limitations

There are several limitations to this analysis:

- **I used an existing Kaggle dataset rather than collecting data directly from students or schools.**
- **The analysis identified relationships and differences but did not establish cause and effect.**
- **The findings may not apply to every student population or educational setting.**
- **The analysis was limited to the variables available in the dataset and the methods used in Excel.**

- **Further analysis and additional data would be needed to understand other influences on student performance and evaluate the effectiveness of potential interventions.**

7. Conclusion

Through this project, I used Excel to investigate student performance by examining five factors: study time, internet access, part-time employment, extracurricular activities, and parental education. I found that the observed relationships and differences varied across these factors. Internet access and part-time employment showed more noticeable differences in average scores, while the differences across extracurricular activities and parental education groups were relatively small.

Overall, my analysis led me to conclude that student performance should be examined in relation to a combination of factors rather than assuming that one factor alone explains the results. The project also helped me practise moving from raw data and individual comparisons to interpreting findings, communicating results through a dashboard, and developing recommendations while acknowledging the limitations of the analysis.